

OS LAB 4 TASKS

TASK#1).

Ubuntu 1 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
new-user@new-user-VirtualBox: ~  
new-user@new-user-VirtualBox:~$ gcc Q1.c -o program1  
Q1.c: In function 'main':  
Q1.c:18:5: error: 'c2' undeclared (first use in this function); did you mean 'f2'  
'?  
18 |     c2 = fork();  
    |     ^~  
    |     f2  
Q1.c:18:5: note: each undeclared identifier is reported only once for each function it appears in  
Q1.c:31:1: error: expected declaration or statement at end of input  
31 | }  
    | ^  
new-user@new-user-VirtualBox:~$ nano Q1.c  
new-user@new-user-VirtualBox:~$ gcc Q1.c -o program1  
Q1.c: In function 'main':  
Q1.c:31:1: error: expected declaration or statement at end of input  
31 | }  
    | ^  
new-user@new-user-VirtualBox:~$ nano Q1.c  
new-user@new-user-VirtualBox:~$ gcc Q1.c -o program1  
new-user@new-user-VirtualBox:~$ ./program1  
Child Pid 3555  
Child PPID 3554  
Parent Finished new-user@new-user-VirtualBox:~$
```

```

new-user@new-user-VirtualBox: ~
GNU nano 7.2 Q1.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t f1 = fork();
    pid_t c2;

    if(f1 == 0)
    {
        printf("Child Pid %d\n",getpid());
    }
    else
    {
        c2 = fork();
        if(c2 == 0)
        {
            printf("Child PPID %d\n",getppid());
        }
        else
        {
            wait(NULL);
            wait(NULL);
            printf("Parent Finished ");
        }
    }
}

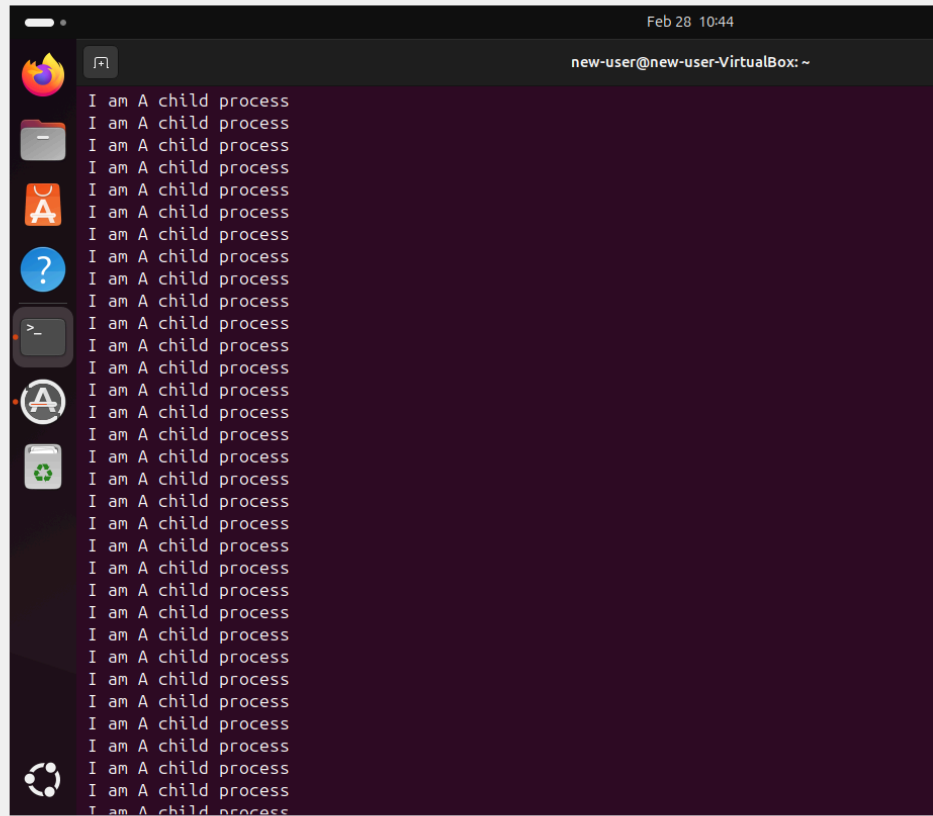
```

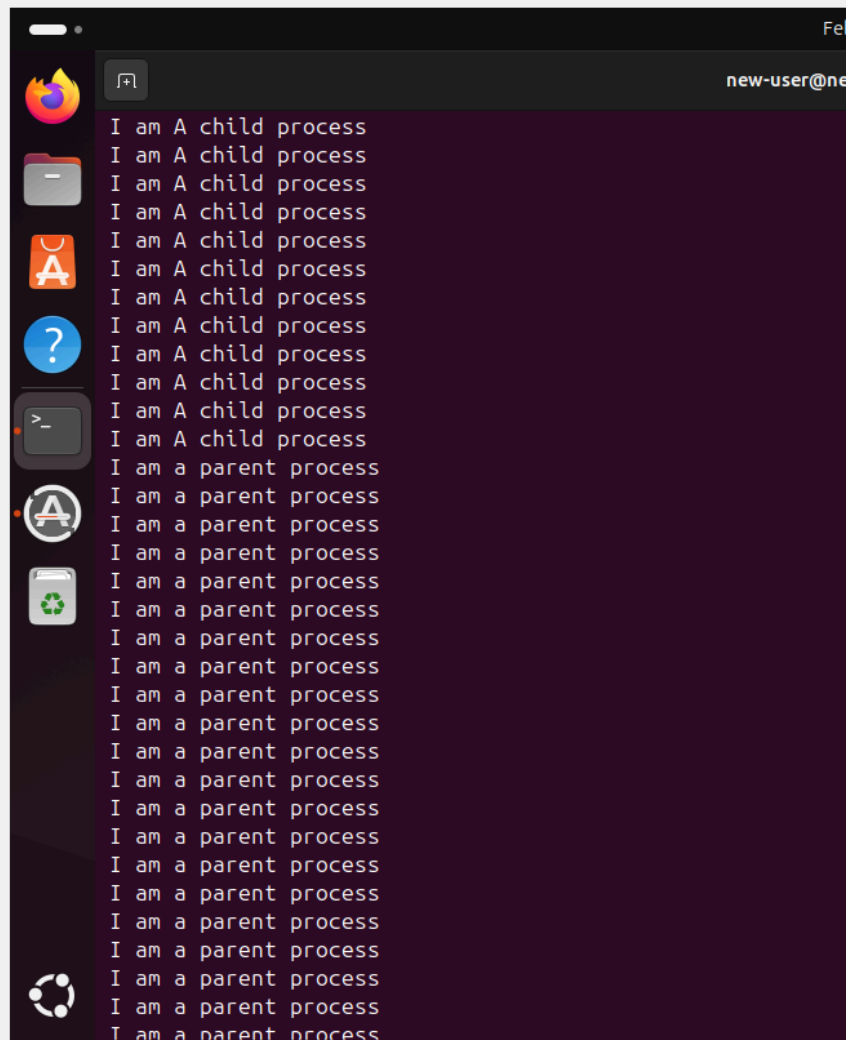
TASK#2).

 Ubuntu 1 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

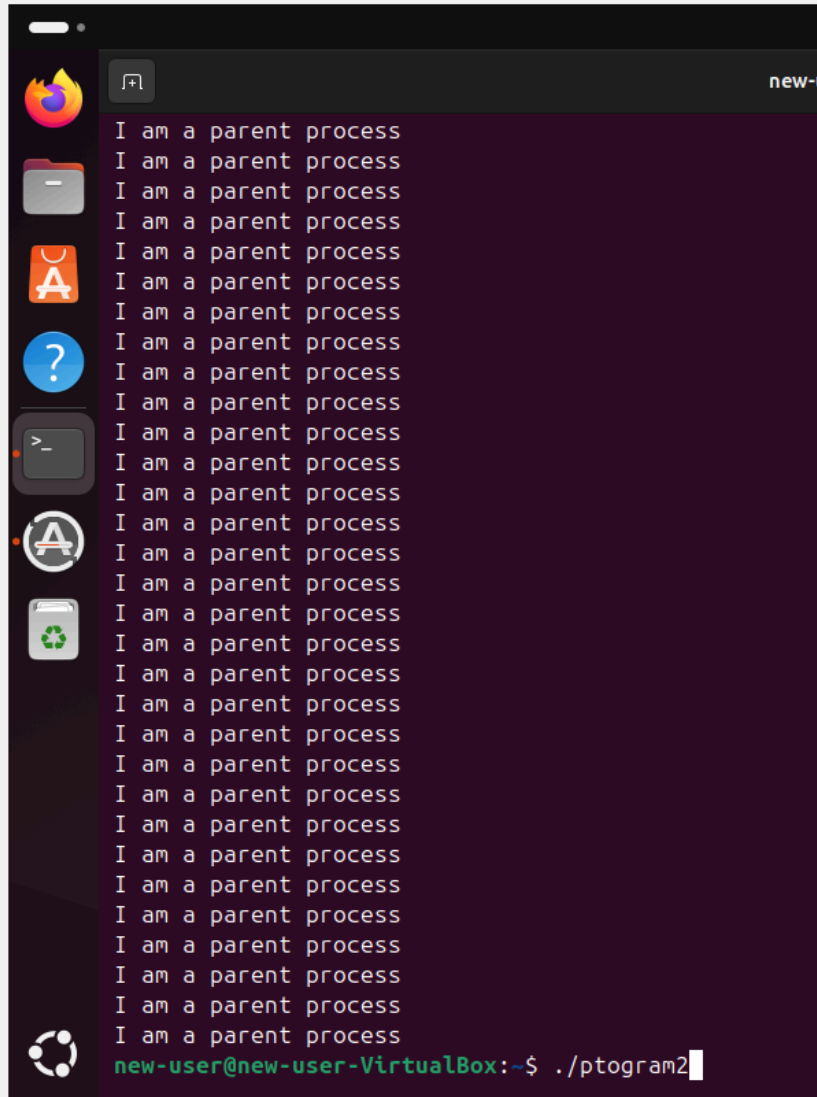
[illegible]





Ubuntu 1 [Running] - Oracle VirtualBox

File Machine View Input Devices Help



TASK#3).

Ubuntu 1 [Running] - Oracle VirtualBox

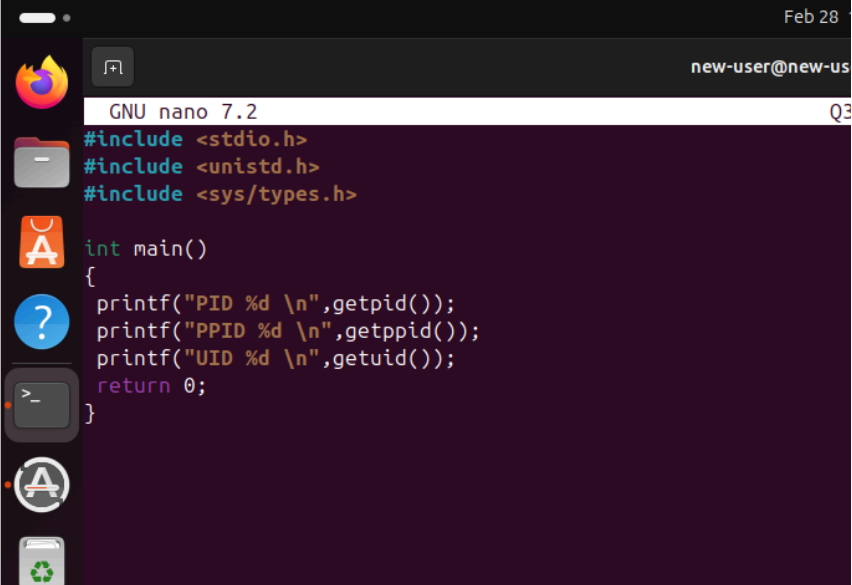
File Machine View Input Devices Help

A screenshot of a Linux terminal window titled "new-user@new-user-VirtualBox". The terminal shows a series of "I am a parent process" messages, followed by commands to compile a program named Q3.c. The first compilation attempt fails with a "fatal error: unist.h: No such file or directory". After correcting the include directive to <unistd.h>, the second compilation attempt fails with "cannot find prog3: No such file or directory". Finally, running the executable ./prog3 shows its PID (3716), PPID (2792), and UID (1000).

```
new-user@new-user-VirtualBox:~$ nano Q3.c
new-user@new-user-VirtualBox:~$ gcc Q3.c prog3
Q3.c:2:10: fatal error: unist.h: No such file or directory
   2 | #include <unist.h>
     |           ^~~~~~
compilation terminated.
new-user@new-user-VirtualBox:~$ nano Q3.c
new-user@new-user-VirtualBox:~$ gcc Q3.c prog3
/usr/bin/ld: cannot find prog3: No such file or directory
collect2: error: ld returned 1 exit status
new-user@new-user-VirtualBox:~$ gcc Q3.c -o prog3
new-user@new-user-VirtualBox:~$ ./prog3
PID 3716
PPID 2792
UID 1000
new-user@new-user-VirtualBox:~$
```

Ubuntu 1 [Running] - Oracle VirtualBox

File Machine View Input Devices Help



The screenshot shows a terminal window titled "new-user@new-us" with a date of "Feb 28". The terminal is running the GNU nano 7.2 text editor. The code being edited is a C program that includes `<stdio.h>`, `<unistd.h>`, and `<sys/types.h>`. The `main` function prints the process ID (PID), parent process ID (PPID), and user ID (UID) using `getpid()`, `getppid()`, and `getuid()` respectively, and then returns 0.

```
GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>

int main()
{
    printf("PID %d \n",getpid());
    printf("PPID %d \n",getppid());
    printf("UID %d \n",getuid());
    return 0;
}
```

TASK#4).


```
new-user@new-user-VirtualBox: ~  
new-user@new-user-VirtualBox:~$ nano input.txt  
new-user@new-user-VirtualBox:~$ nano Q5.c  
new-user@new-user-VirtualBox:~$ gcc Q5.c -o pro5  
new-user@new-user-VirtualBox:~$ ./pro5  
File Succesfully Opened  
new-user@new-user-VirtualBox:~$ cat output.txt  
Hello World!..  
new-user@new-user-VirtualBox:~$ nano Q5.c  
new-user@new-user-VirtualBox:~$ nano Q4.c  
new-user@new-user-VirtualBox:~$ gcc Q4.c -o pro4  
Q4.c: In function 'main':  
Q4.c:42:22: error: 'j' undeclared (first use in this function)  
    42 |         sum += arr[j];  
       |                     ^  
Q4.c:42:22: note: each undeclared identifier is reported only once for each function it appears in  
new-user@new-user-VirtualBox:~$ nano Q4.c  
new-user@new-user-VirtualBox:~$ gcc Q4.c -o pro4  
new-user@new-user-VirtualBox:~$ ./pro4  
Usage ./pro4 num1 num2 num3  
new-user@new-user-VirtualBox:~$ nano Q4.c  
new-user@new-user-VirtualBox:~$ gcc Q4.c -o pro4  
new-user@new-user-VirtualBox:~$ ./pro4 10 20 30 40  
Sum 100  
Average is 25.000000  
^C  
new-user@new-user-VirtualBox:~$ nano Q4.c  
new-user@new-user-VirtualBox:~$ gcc Q4.c -o pro4  
new-user@new-user-VirtualBox:~$ ./pro4 10 20 30 40  
Sum 100  
Maximum is 40  
Average is 25.00  
new-user@new-user-VirtualBox:~$
```

```

GNU nano 7.2 Q4.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main(int argc, char *avg[])
{
    if(argc < 2)
    {
        printf("Usage %s num1 num2 num3 \n",avg[0]);
        return 1;
    }

    int n = argc -1;
    int arr[n];

    for(int i =0;i<n;i++)
    {
        arr[i] = atoi(avg[i+1]);
    }

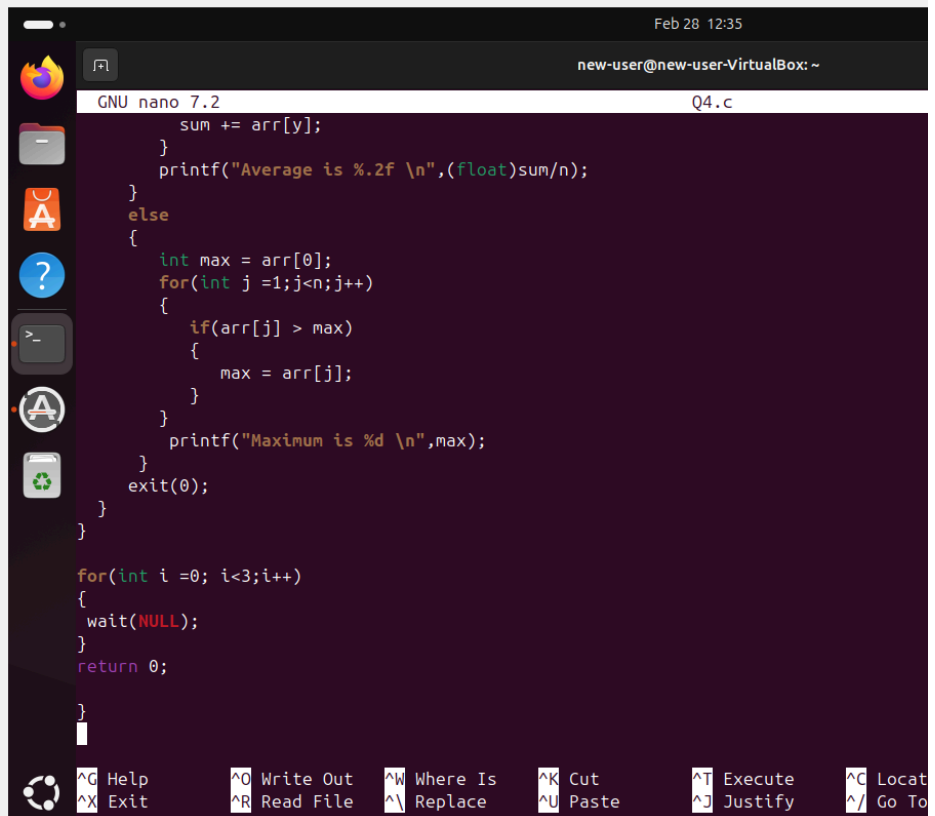
    for(int i =0; i<3;i++)
    {
        pid_t p1 = fork();
        if(p1 == 0)
        {
            if(i == 0)
            {

```

```

GNU nano 7.2 Q4.c
int sum =0;
for(int j =0; j<n;j++)
{
    sum += arr[j];
}
printf("Sum %d \n",sum);
}
else if(i == 1)
{
    int sum =0;
    for(int y =0;y<n;y++)
    {
        sum += arr[y];
    }
    printf("Average is %.2f \n",(float)sum/n);
}
else
{
    int max = arr[0];
    for(int j =1;j<n;j++)
    {
        if(arr[j] > max)
        {
            max = arr[j];
        }
    }
    printf("Maximum is %d \n",max);
}

```



TASK#5).

Ubuntu 1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

```
Feb 28 12:17
new-user@new-user-VirtualBox: ~
new-user@new-user-VirtualBox:~$ nano Q5.c
new-user@new-user-VirtualBox:~$ gcc Q5.c -o pro5
Q5.c:2:10: fatal error: fcnti.h: No such file or directory
  2 | #include <fcnti.h>
    |             ^~~~~~
compilation terminated.
new-user@new-user-VirtualBox:~$ nano Q5.c
new-user@new-user-VirtualBox:~$ gcc Q5.c -o pro5
Q5.c: In function 'main':
Q5.c:27:41: error: expected expression before '>' token
  27 | while(byte = read(fd1,buffer,BUFFER))>0)
    |                                         ^
Q5.c:27:43: error: expected statement before ')' token
  27 | while(byte = read(fd1,buffer,BUFFER))>0)
    |                                         ^
Q5.c:29:22: error: 'bytes' undeclared (first use in this function); did you mean 'byte'?
  29 | write(fd2,buffer,bytes);
    |                     ^~~~~
    |                     byte
Q5.c:29:22: note: each undeclared identifier is reported only once for each function it appears in
new-user@new-user-VirtualBox:~$ nano Q5.c
new-user@new-user-VirtualBox:~$ gcc Q5.c -o pro5
new-user@new-user-VirtualBox:~$ ./pro5
Error Opening in File: No such file or directory
new-user@new-user-VirtualBox:~$ nano input.txt
new-user@new-user-VirtualBox:~$ nano Q5.c
new-user@new-user-VirtualBox:~$ gcc Q5.c -o pro5
new-user@new-user-VirtualBox:~$ ./pro5
File Succesfully Opened
new-user@new-user-VirtualBox:~$ cat output.txt
Hello World!
new-user@new-user-VirtualBox:~$
```

```
GNU nano 7.2 Q5.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

#define BUFFER 1024

int main()
{
    int fd1 = open("Input.txt",O_RDONLY);
    if(fd1 < 0)
    {
        perror("Error Opening in File");
        exit(1);
    }

    int fd2 = open("output.txt",O_WRONLY | O_CREAT | O_TRUNC,0644);
    if(fd2 < 0)
    {
        perror("Error in opening file\n");
        exit(1);
    }

    char buffer[BUFFER];
    int byte;

    while((byte = read(fd1,buffer,BUFFER))>0)
    {
        Read 37 lines
    }
}
```

```
GNU nano 7.2 Q5.c
if(fd1 < 0)
{
    perror("Error Opening in File");
    exit(1);
}

int fd2 = open("output.txt",O_WRONLY | O_CREAT | O_TRUNC,0644);
if(fd2 < 0)
{
    perror("Error in opening file\n");
    exit(1);
}

char buffer[BUFFER];
int byte;

while((byte = read(fd1,buffer,BUFFER))>0)
{
    write(fd2,buffer,byte);
}

close(fd1);
close(fd2);
printf("File Succesfully Opened\n");
return 0;
}
```

TASK#6).

Ubuntu 1 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
Feb 28 11:52
new-user@new-user-VirtualBox: ~
new-user@new-user-VirtualBox:~$ nano Q7.
new-user@new-user-VirtualBox:~$ nano Q7.c
new-user@new-user-VirtualBox:~$ gcc Q7.c -o pro7
new-user@new-user-VirtualBox:~$ ./pro7
Desktop  gb.sh      h.sh      Pictures  program1  Q1.c  Q5.c  t2.sh  t5.sh  Videos
Documents gh1.sh      lab3.sh   pro7      program2  Q2.c  Q7.c  t3.sh  t6.sh
Downloads gh.sh       Music     prog3     ptogram2  Q3.c  sample t3..sh t7.sh
g2.sh     hello.sh.save os_lab_24k_0559 progr4    Public   Q4.c  snap  t4.sh  Templates
Child Executednew-user@new-user-VirtualBox:~$ nano Q7.c
new-user@new-user-VirtualBox:~$ nano Q6.c
new-user@new-user-VirtualBox:~$ gcc Q6.c -o pro6
new-user@new-user-VirtualBox:~$ ./pro6
Desktop  gb.sh      h.sh      Pictures  progr4    Public  Q4.c  sample t3..sh  t7.sh
Documents gh1.sh      lab3.sh   pro6      program1  Q1.c  Q5.c  snap  t4.sh  Templates
Downloads gh.sh       Music     pro7      program2  Q2.c  Q6.c  t2.sh  t5.sh  testdir
g2.sh     hello.sh.save os_lab_24k_0559 prog3     ptogram2  Q3.c  Q7.c  t3.sh  t6.sh  Videos
cp: cannot stat 'input.txt': No such file or directory
mkdir: failed to remove 'testDir': No such file or directory
new-user@new-user-VirtualBox:~$
```

```

GNU nano 7.2                                Q6.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    if(fork()==0)
    {
        execlp("mkdir","mkdir","testdir",NULL);
    }

    wait(NULL);

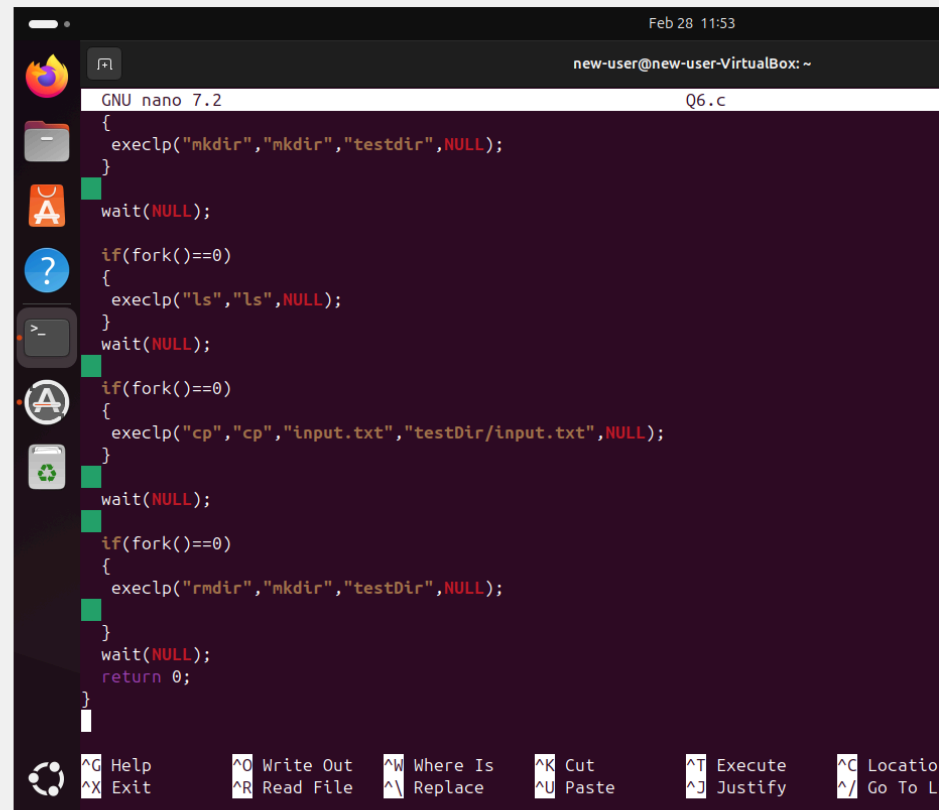
    if(fork()==0)
    {
        execlp("ls","ls",NULL);
    }
    wait(NULL);

    if(fork()==0)
    {
        execlp("cp","cp","input.txt","testdir/input.txt",NULL);
    }

    wait(NULL);

    if(fork()==0)
    {

```

The screenshot shows a terminal window titled "new-user@new-user-VirtualBox: ~" with a date and time of "Feb 28 11:53". The terminal is running the GNU nano 7.2 text editor, editing a file named "Q6.c". The code in the file is a C program that uses `fork()` to create three child processes. Each child process performs a different task: the first creates a directory, the second lists the directory, and the third copies a file. All processes wait for each other using `wait(NULL)` before returning 0. The terminal has a dark purple background with a sidebar on the left containing icons for file manager, application store, help, terminal, and other utilities. At the bottom, there is a status bar with various keyboard shortcuts for nano editor functions like Help, Exit, Write Out, Read File, Where Is, Replace, Cut, Paste, Execute, Justify, Location, and Go To Line.

```
GNU nano 7.2 Q6.c
{
    execlp("mkdir","mkdir","testdir",NULL);
}

wait(NULL);

if(fork()==0)
{
    execlp("ls","ls",NULL);
}
wait(NULL);

if(fork()==0)
{
    execlp("cp","cp","input.txt","testDir/input.txt",NULL);
}

wait(NULL);

if(fork()==0)
{
    execlp("rmdir","rmdir","testDir",NULL);
}

wait(NULL);
return 0;
}
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

TASK#7).

Ubuntu 1 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
new-user@new-user-VirtualBox: ~  
new-user@new-user-VirtualBox:~$ nano Q7.  
new-user@new-user-VirtualBox:~$ nano Q7.c  
new-user@new-user-VirtualBox:~$ gcc Q7.c -o pro7  
new-user@new-user-VirtualBox:~$ ./pro7  
Desktop  gb.sh      h.sh      Pictures  program1  Q1.c  Q5.c  t2.sh  t5.sh  Videos  
Documents gh1.sh    lab3.sh   pro7      program2  Q2.c  Q7.c  t3.sh  t6.sh  
Downloads gh.sh     Music     prog3     ptogram2  Q3.c  sample t3..sh t7.sh  
g2.sh    hello.sh.save os_lab_24k_0559 progr4    Public  Q4.c  snap  t4.sh  Templates  
Child Executednew-user@new-user-VirtualBox:~$
```

Ubuntu 1 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
GNU nano 7.2 Q7.c  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/wait.h>  
  
int main()  
{  
    pid_t pid = fork();  
    if(pid == 0)  
    {  
        execlp("ls","ls",NULL);  
    }  
    else  
    {  
        wait(NULL);  
        printf("Child Executed");  
    }  
    return 0;  
}
```

Read 19 lines

Help Write Out Where Is Cut Execute Location W-U Undo W-A Set Mark
Exit Read File Replace Paste Justify Go To Line W-E Redo W-G Copy